AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

ANNUAL EXAMINATIONS (THEORY) 2025

General Mathematics Paper II

Time: 1 hour 40 minutes Marks: 30

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL the Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a simple calculator if you wish.

List of Formulae

Note:

All symbols used in the formulae have their usual meaning.

Financial Mathematics

$$I = PT \times \frac{R}{100}$$

Basic Statistics

$$\overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{n} \quad \text{or} \quad \overline{X} = \frac{\sum fx}{\sum f}$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{Standard deviation} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Quadratic Equations

$$ax^2 + bx + c = 0,\ a \neq 0 \qquad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \text{Disc} = b^2 - 4ac$$

Algebraic Manipulation

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a-b)^2 = a^2 - 2ab + b^2 \qquad a^2 - b^2 = (a+b)(a-b)$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \qquad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Arithmetic and Geometric Sequence

$$a_n = a_1 + (n-1)d \qquad AM = \frac{a+b}{2} \qquad a_n = a_1 r^{n-1} \qquad GM = \pm\sqrt{ab}$$

Area and Volumes

$$\text{Area of a circle} = \pi r^2 \qquad \text{Area of a triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Volume of a cube} = l^3 \qquad \text{Volume of a cuboid} = l \times b \times h \qquad \text{Volume of a sphere} = \frac{4}{3} \times \pi r^3$$

$$\text{Volume of a cone} = \frac{1}{3} \times \pi r^2 \times h$$

$$\text{Volume of a cylinder} = \pi r^2 \times h$$

Introduction to Coordinate Geometry

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

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(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 3 Marks)

a. A dealer sells a car at cash price of Rs 1,500,000. If Asif buys this car on instalments, then he would need to pay 15% down payment and the remaining amount in monthly instalments of Rs 60,000 for 2 years.

In the given situation, calculate the

i. initial deposit. (1 Mark)

ii. price on installments. (1 Mark)

iii. additional amount after paying down payment and monthly installments. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.1.)

b. Sharif earns Rs 150,000 as monthly gross salary. His salary break up is tabulated as given.

Basic Salary (Rs)	Medical Allowance (Rs)	Food Allowance (Rs)	Professional Allowance (Rs)
110,000	15,000	15,000	10,000

If the income tax rate is 12.5%, then calculate the

i. monthly income tax. (1 Mark)

ii. monthly consumable income. (1 Mark)

iii. gross annual income. (1 Mark)

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Q.2. (Total 4 Marks)

On a certain day, 9 friends spend the following amount (in Rs) on a shopping trip.

3800, 2500, 1200, 1500, 1700, 2000, 1500, 3200, 3500

i. Find the median amount that was spent. (2 Marks)

ii. If the mean of 5 numbers 18, k, 45, 30 and 8 is 25, then find the value of k. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. It is given that $p(x) = x^2 - 3x + 2$ and $q(x) = (x-1)^2$ are two polynomials. For the given polynomials, find the LCM and HCF.

b. Find the square root of the expression $x^4 - 2x^3 - x^2 + 2x + 1$ by division method.

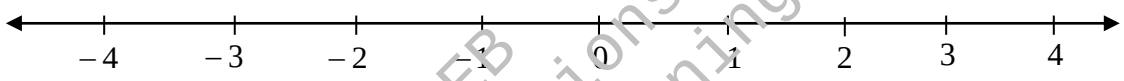
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Q.4. (Total 3 Marks)

If $x \in R$, then for $-2x+3 \geq -1$,

i. solve for *x*. (2 Marks)

ii. show the solution on the given number line. (1 Mark)



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Q.5. (Total 4 Marks)

The product of two consecutive positive odd integers is 1443. Find the smaller of the two integers.

Q.6. (Total 4 Marks)

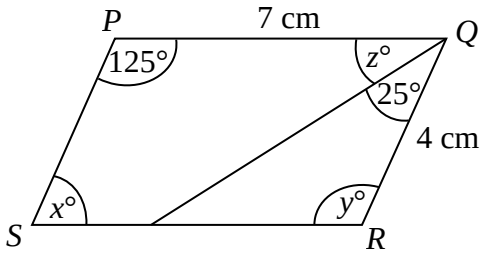
If A_1, A_2, A_3, and A_4 are arithmetic means between 2 and 22, then find the value of A_3.

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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a. In the given diagram, the quadrilateral *PQRS* is a parallelogram.



NOT TO SCALE

For the given parallelogram, find the

i. angle $y°$. (1 Mark)

ii. angle $x°$. (1 Mark)

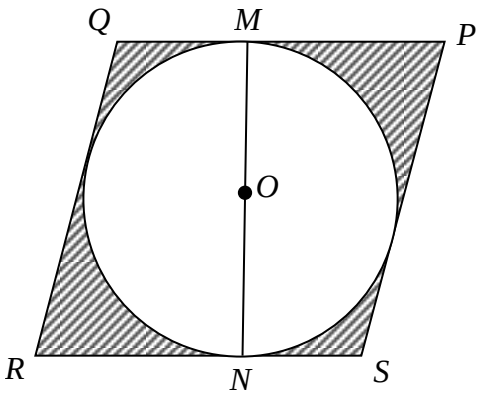
iii. angle $z°$. (1 Mark)

iv. length of *SR*. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

b. In the given diagram, a circle with centre *O* is given. *MN* is the diameter of the circle and it is perpendicular to *RS*.



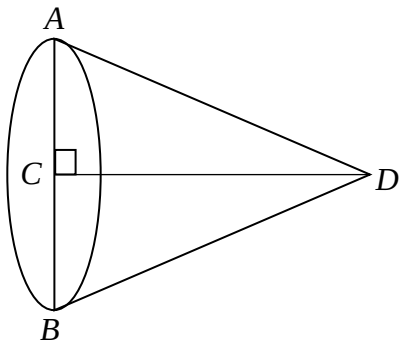
NOT TO SCALE

If *RS* = 6.5 cm, *QP* = 6.5 cm and *MN* = 6 cm, then calculate the area of the shaded region.

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Q.8. (Total 4 Marks)

The given diagram shows a right circular cone.



NOT TO SCALE

Calculate the diameter of the cone if its height is 12 inches and volume is 616 cubic inches.

(**Note**: Take $\pi = \frac{22}{7}$)

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END OF PAPER

Please use this page for rough work

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